Database Standards

ISIN 325 - Database Security

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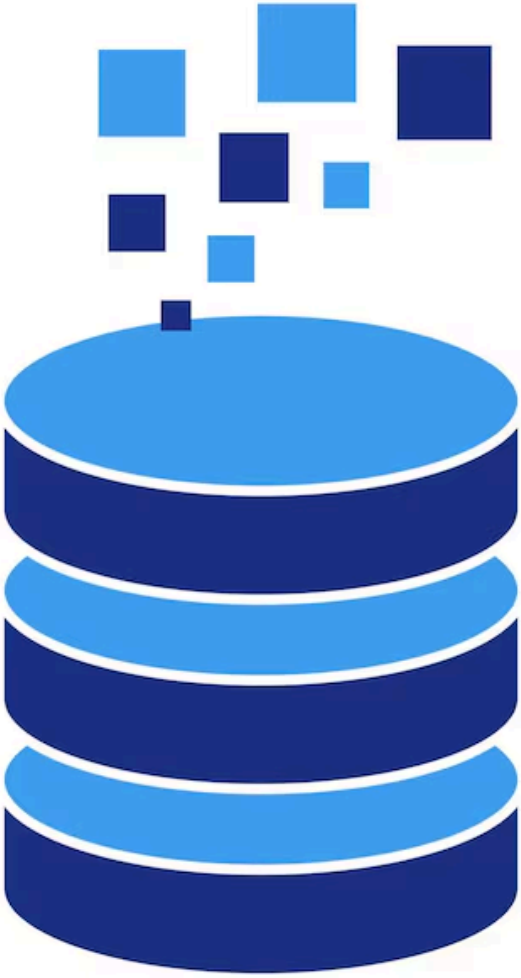

Overview: In this assignment for ISIN 325, we were tasked with building a data dictionary of the Meijer database and creating a standards document.

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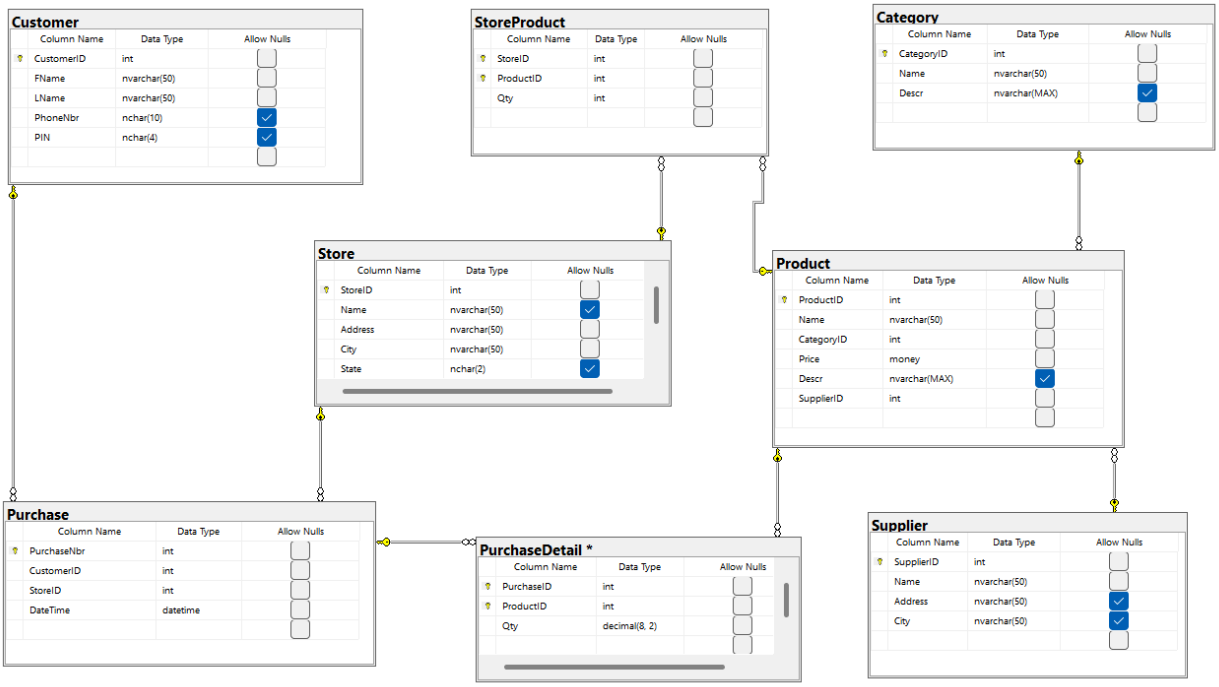
Database Design



Tables

Category Table: This table has records on the categories of all the products in the Meijer database. It includes data on the description and name of the category.

Column Name	Nullable	Data Type	Max Length	Example of data
CategoryID	NO	int	NULL	32
Desc	YES	nvarchar	MAX	Health products
Name	NO	nvarchar	50	Health

Customer Table: This table has information on all customer data. It has records on the customer's name, phone number, and PIN.

Column Name	Nullable	Data Type	Max Length	Example of data
CustomerID	NO	int	NULL	92
FName	NO	nvarchar	50	Jessica
LName	NO	nvarchar	50	Smith
PhoneNbr	YES	nchar	10	5177635459
YES	YES	nchar	4	1234

Product Table: The product table contains information such as product category, price, a description, and the supplier of the product.

Column Name	Nullable	Data Type	Max Length	Example of data
Name	NO	nvarchar	50	Tylenol
CategoryID	NO	int	NULL	13
ProductID	NO	INT	NULL	1
Price	NO	money	NULL	$45
Descr	YES	nvarchar	MAX	medication for headache
SupplierID	NO	int	NULL	2

Purchase Table: This table gathers information on the date, location, and customer for every purchase.

Column Name	Nullable	Data Type	Max Length	Example of data
PurchaseNbr	NO	int	NULL	97
CustomerID	NO	int	NULL	3
StoreID	NO	int	NULL	4
DateTime	NO	datetime	NULL	45423

PurchaseDetail Table: This is a bridge table that uses the purchase and product tables to identify products and the quantity of every transaction.

Column Name	Nullable	Data Type	Max Length	Example of data
PurchaseNbr	NO	int	NULL	843
ProductID	NO	int	NULL	645
Qty	NO	Decimal	8,2	3

Store Table: This table stores the name, address, state, and zip code of every Mejier location.

Column Name	Nullable	Data Type	Max Length	Example of data
Name	YES	nvarchar	50	Howell
StoreID	NO	int	NULL	156
Address	NO	nvarchar	50	785 Campus Dr.
State	YES	nchar	2	MI
ZipCode	NO	nchar	10	48892

Supplier Table: This table records every supplier for the products Meijer carries and stores the name, address, and city.

Column Name	Nullable	Data Type	Max Length	Example of data
Name	NO	nvarchar	50	Factory Three
Address	YES	nvarchar	50	123 E Supplier St
SupplierID	NO	int	NULL	15
City	YES	nvarchar	50	Big Rapids

StoreProduct Table: This table compiles all of the products in each store as well as the quantity of the product.

Column Name	Nullable	Data Type	Max Length	Example of data
StoreID	NO	int	NULL	123
ProductID	NO	int	NULL	156
Qty	NO	int	NULL	13

Standards

The organization will follow these standards to ensure the database is equal throughout.

- Follows camel case for all databases, tables, and fields
- All records must be in lower case except for the first letter of sentences and words
- Names must not have underscores (_)
- Limit of 25 characters per title

Security

The organization will follow these security standards to ensure the database is protected.

- Use stored procedures for applications and other tasks to limit security risk and run queries faster
- Implement role-based access control
- Ensure all employees who access the database have separate logins
- Enforce strict logging of all database activity

Summary

This Meijer database uses consistent standards and appropriate security controls, keeping the company safe and organized. The data dictionary also serves as a reference for any employees who need to examine the types of data allowed in each table for analysis and decision-making.